COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 | Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	Sep 15, 2026	**DATE SUBMITTED:**	Sep 17, 2026
NAME:	Dela Cruz, Jared Isaiah Santos, Paterno Miguel Serrano, Gabriel Immanuel Urbano, Vince Andrew Vizcayno, Ellaiza	**INSTRUCTOR:**	Engr. Jimlord Quejado

ASSIGNMENT NO. 9.1 Binary Search Trees

1. PROCEDURE OUTPUT

Summarized Answers	
Question	1. What is a binary search tree?
Answer	A binary search tree is a data structure where each node has at most two children. Every value in the left subtree is smaller than its parent node, and every value in the right subtree is bigger. This ordering is what makes searching, adding, and removing data fast.

Question	2. Where can binary search tree be used?
Answer	Binary search trees are used wherever data needs to be searched quickly instead of checking every item one by one. This includes database indexing for large sets of records, search engines, and file systems that need to organize and locate files efficiently.

Question	3. What is a tree traversal?
Answer	Tree traversal is the process of visiting every node in a tree exactly once, in a specific order. Since a tree branches out instead of being a straight line, traversal gives the computer a clear path to follow so it processes each node without skipping any or repeating one.

Question	4. Differentiate a post-order and pre-order traversal give examples.
Answer	The difference between pre-order and post-order traversal is when the root node gets visited. In pre-order, you visit the root first, then the left subtree, then the right subtree. In post-order, you visit the left subtree first, then the right subtree, and visit the root last. For example, in a tree with root 1, left child 2, and right child 3: pre-order gives 1, 2, 3, while post-order gives 2, 3, 1.

Question	5. What is a parse tree and its use?
Answer	A parse tree is a diagram that breaks down an expression, sentence, or piece of code into its grammatical or structural parts based on a set of rules. It is used to show how the different parts of the expression fit together, which helps compilers or programs check and evaluate it correctly.

Individual Answers:	
Dela Cruz	**1. What is a binary search tree?** A binary search tree is a way of organizing information in a computer so you can find, add, or remove data fast. The key rule that makes it work is simple. Every value on the left side must be smaller than the middle node, and every value on the right side must be bigger. **2. Where can binary search tree be used?** Binary search trees are heavily used in computer systems whenever you need to look up data quickly without checking every single item one by one. Databases use them to index huge amounts of user records so queries finish in a fraction of a second. They are also used in computer file systems to organize folders and files. **3. What is a tree traversal?** Tree traversal means visiting every single node in a tree structure exactly once to read or work with its data. Because tree data branches out in different directions instead of sitting in a straight line, your computer needs a clear sequence to follow. Traversal provides that systematic order so the program covers every node without skipping anything or visiting the same spot twice. **4. Differentiate a post-order and pre-order traversal give examples.** In pre-order traversal, you deal with the parent node first, then go to the left branch, and end with the right branch. In post-order traversal, you work through the left and right branches first, and save the parent node for last. **5. What is a parse tree and its use?** A parse tree is a branching diagram that breaks down a piece of text or code into its basic grammatical parts. When you write a program or input a math expression, the system builds a parse tree to map out how all the words, numbers, and symbols fit together.
Santos	**1. What is a Binary Search Tree?** It is a type of tree data structure where each node has two children with values in the left branch being smaller than the node, while the values in the right subtree are larger. **2. Where can Binary Search Trees be used?** It can be used for searching data and storing data. They are useful when data needs to be kept organized and easily searchable.

	3. What is a tree traversal? It is a method of processing every node in a tree in a specific way. **4. Differentiate a post-order and pre-order traversal. Give examples.** The main difference between pre-order and post-order traversal is when the root node is visited. In pre-order traversal, the root is visited first. While in post-order traversal, the root is visited last. **5. What is a parse tree and its use?** A tree that represents the structure of an expression or sentence according to a set of rules or grammar.
Serrano	**1. What is a binary search tree?** A binary search tree is a tree data structure where each node has only two children. All the number that is smaller value is placed on the left while the higher values is on the right **2. Where can binary search tree be used?** A binary search tree can be used for searching data in a large databases that need sorted data**.** **3. What is a tree traversal?** Tree traversal is the process of visiting or accessing every node in a tree in a specific order. **4. Differentiate a post-order and pre-order traversal. Give examples.** Pre-order visits the root firsts, then by the left and right subtrees, while the post-order visits the left and right first, then to the root. **5. What is a parse tree and its use?** A parse tree is a tree that shows the grammatical breakdown of a string of text into its different parts. It is used to represent the structure of sentences or expressions and help analyze them.
Urbano	**1. What is a binary search tree?** A binary search tree is a tree data structure where each parent node can have two child nodes. All numbers on the left are smaller than the parent node and all numbers on the right are bigger. **2. Where can binary search tree be used?** Binary search trees are used in search engines to find stored data fast and in

	database systems to look up info easily. **3. What is a tree traversal?** Tree traversal means visiting every single node in a tree one by one in a specific order so you can print the data inside them. **4. Differentiate a post-order and pre-order traversal give examples.** In pre-order traversal you check the root node first, then go to the left side, and lastly go to the right side. In post-order traversal you check the left side first, then the right side, and lastly you check the root node. **5. What is a parse tree and its use?** A parse tree is a diagram shaped like a tree that shows how a line of code works. Compilers use it to check if code is written correctly.
Vizcayno	**1. What is a binary search tree?** A binary search tree is a data structure where all nodes in the left subtree contain values less than the root or parent node, while all nodes in the right subtree contain values greater than the root or parent node. **2. Where can binary search trees be used?** They can be used for efficiently storing and searching data, such as in databases and other applications that require fast searching. **3. What is a tree traversal?** A tree traversal is the process of going through the nodes of a tree by visiting the root node and its left and right subtrees in a specific order. **4. Differentiate a post-order and pre-order traversal. Give examples.** In pre-order traversal, you first visit the root, then explore the left subtree, and then the right subtree. For example, if the tree has 1 as the root, 2 on the left, and 3 on the right, the traversal is 1, 2, 3. In post-order traversal, you first traverse the left subtree, then the right subtree, and finally visit the root. The traversal would be 2, 3, 1. **5. What is a parse tree and its use?** A parse tree breaks down an expression into its different parts in a structured way, making it easier to understand and evaluate the expression correctly.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.